

# Project Design Phase

## Problem – Solution Fit Template

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|---------------|--------------------------------------------------------|
| Date          | 18 June 2026                                           |
| Team ID       |                                                        |
| Project Name  | ShopEz (Full Stack MERN Application)                   |
| Student Name  | INAPAKUTHIKA BHARATH VARMA (bharathvarma906@gmail.com) |
| Maximum Marks | 2 Marks                                                |

### Problem – Solution Fit Framework:

A successful Problem-Solution Fit is realized when a proposed system effectively targets confirmed user pain points and behavioral trends. This ensure that ShopEz's feature set is custom-built to resolve specific bottlenecks currently found within the digital retail environment.

### Purpose:

- ☐ Address intricate virtual shopping hurdles for various user groups.
- ☐ Drive engagement by leveraging the high-speed MERN development stack.
- ☐ Align the technical roadmap with user-oriented functional requirements.
- ☐ Cultivate user confidence via secure data handling and fluid order management.
- ☐ Refine the e-commerce experience for the modern digital marketplace.

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| <b>1. TARGET AUDIENCE</b> <small>CS</small> <ul style="list-style-type: none"><li>• Time-constrained professionals.</li><li>• Digital-native students.</li><li>• Parents seeking children's apparel.</li><li>• Independent retail managers.</li></ul>                   | <b>6. USER CONSTRAINTS</b> <small>CC</small> <ul style="list-style-type: none"><li>• Short window for transactions.</li><li>• Variable mobile data connectivity.</li><li>• Preference for intuitive interfaces.</li><li>• High demand for payment privacy.</li></ul>                          | <b>5. CURRENT ALTERNATIVES</b> <small>AS</small> <ul style="list-style-type: none"><li>• Mass-market e-commerce giants.</li><li>• Social media marketplace ads.</li><li>• Physical neighborhood stores.</li><li>• Outdated web platforms.</li></ul>               |
| <b>2. CORE OBJECTIVES</b> <small>J&amp;P</small> <ul style="list-style-type: none"><li>• Swiftly browse specific clothing.</li><li>• Manage carts and orders reliably.</li><li>• Apply filters (size/type/gender).</li><li>• Rapid inventory synchronization.</li></ul> | <b>9. ROOT ISSUES</b> <small>RC</small> <ul style="list-style-type: none"><li>• Overly heavy web frameworks.</li><li>• Session loss in shopping carts.</li><li>• Confusing menu structures.</li><li>• Vulnerable information protocols.</li></ul>                                             | <b>7. OBSERVED HABITS</b> <small>BE</small> <ul style="list-style-type: none"><li>• Exits site if latency is high.</li><li>• Compares prices via multiple tabs.</li><li>• Relies heavily on category filters.</li><li>• Monitors delivery status daily.</li></ul> |
| <b>3. TRIGGERS &amp; 4. STATES</b> <small>TR/EM</small> <ul style="list-style-type: none"><li>• Sudden need for seasonal wear.</li><li>• Trigger: Consistent site speed.</li><li>• Feeling: Rewarded by offers.</li><li>• Concerns: Delivery timelines.</li></ul>       | <b>10. PROPOSED SOLUTION</b> <small>SL</small> <p><b>ShopEz: MERN Architecture</b></p> <ul style="list-style-type: none"><li>• Responsive SPA dynamics.</li><li>• Streamlined Admin CRUD portal.</li><li>• JWT-protected authentication.</li><li>• Live transaction log monitoring.</li></ul> | <b>8. ENGAGEMENT CHANNELS</b> <small>CH</small> <ul style="list-style-type: none"><li>• React-optimized mobile web.</li><li>• Marketing link click-throughs.</li><li>• Internal administrative portal.</li><li>• Digital status notifications.</li></ul>          |

**References:**

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>